BEYONDBOTS: PROJECT REPORT

Detecting Fan-Driven and Paid Coordinated Behavior in Indian Film Social Media

Abstract

Indian cinema discourse on social media platforms is heavily shaped by coordinated campaigns that obscure genuine audience sentiment. These campaigns typically stem from two distinct motivators: star-driven fan-club mobilization (fandom intervention) and undisclosed commercial promotion (paid marketing campaigns). While both manifest as coordinated inauthentic behavior (CIB), they differ fundamentally in intent, linguistic signature, and network architecture. This project introduces **BeyondBots**, a 7-stage text mining and explainable machine learning pipeline designed to ingest social media reviews (CSV, PDF, DOCX, TXT) and automatically distinguish organic user opinions from coordinated fan-club activity and paid PR campaigns. Leveraging dense multilingual representations (MuRIL), hierarchical density clustering (HDBSCAN), stylometric feature engineering, and LLM-in-the-loop classification with SHAP surrogate explanations, BeyondBots provides a robust, explainable framework for platform trust and safety teams, entertainment journalists, and advertising regulators to analyze astroturfing campaigns in code-mixed languages.

1. Literature Review & Related Work

Detecting inauthentic coordination on social media requires synthesizing methodologies from duplicate text clustering, astroturfing detection, network community profiling, and NLP stylistic classification. In this review, we examine 15 foundational works across these domains, detailing how their insights shape the implementation of the BeyondBots pipeline.

- **Ratnakar & Kumar (2020)** (Astroturfing Detection in Review Systems): Analyzes patterns in e-commerce reviews, showing that commercial review spam relies heavily on repetitive superlatives and generic language. In BeyondBots, this study informs our 'marketing-copy vocabulary' features, which flags generic praise lacking film-specific reference points.
- **Ferrara et al. (2016)** (The Rise of Social Bots): Argues that bot detection tools relying solely on account age or high frequency fail against human-led networks. This motivates our network community and near-duplicate text cluster approach, as fan-club users are real, long-standing human accounts.
- **Benevenuto et al. (2010)** (Detecting Spammers on Twitter): Focuses on social network interaction topologies (replies, retweets) to isolate spammers. This guides Stage 3 of our pipeline, where we use reply network mapping to uncover coordinated fan-club networks interacting with stars.

- **Devlin et al. (2018)** (BERT: Deep Bidirectional Transformers): Introduces bidirectional contextual embeddings. This forms the mathematical basis for Stage 2, allowing semantic representation of reviews prior to similarity clustering.
- **Khanuja et al. (2021)** (MuRIL: Multilingual Representations for Indian Languages): Develops a multilingual transformer pretrained on code-mixed Indian languages. This is crucial for BeyondBots, as Indian cinema reviews extensively use Hindi-English, Tamil-English, and Telugu-English code-mixing (Hinglish/Tanglish).
- **Campello et al. (2013)** (HDBSCAN Hierarchical Density Clustering): Presents a density-based clustering algorithm that handles variable density clusters and labels outliers as noise. In BeyondBots, this allows near-duplicate posts to group into tight coordinated templates while leaving organic reviews in the noise (-1) bucket.
- **Traag et al. (2019)** (From Louvain to Leiden Community Detection): Introduces the Leiden algorithm for uncovering well-connected network communities. We use this in Stage 3 to identify highly connected, insular subgraphs representing star fan-club networks.
- **Lundberg & Lee (2017)** (SHAP: A Unified Approach to Model Interpretability): Proposes SHAP (SHapley Additive exPlanations) values to interpret models. We train a local Logistic Regression surrogate on our pipeline's stylistic features to match LLM labels, using SHAP to expose global feature importances directly in the Streamlit UI.
- **Pennebaker et al. (2015)** (LIWC: Linguistic Inquiry and Word Count): Demonstrates that stylistic pronouns, honorifics, and pronouns reveal underlying psychology. This informs Stage 4, where we engineered fan-identity metrics like star-honorific counts and community-building pronouns (we/our/us).
- **Blei et al. (2003)** (Latent Dirichlet Allocation (LDA)): Introduces LDA topic modeling. We implement this in Stage 5 to automatically extract thematic keywords for each duplicate cluster, helping summarize the core talking points of a coordinated campaign.
- **Grootendorst (2022)** (BERTopic: Neural Topic Modeling): Details cluster keyword extraction using class-based TF-IDF (c-TF-IDF). This inspired our Stage 5 representation centroid-based keyword summarizer to name clusters for the LLM classifier.
- **Cresci et al. (2017)** (The Paradigm Shift of Social Spambots): Shows that modern spambots act collectively as groups (CIB) rather than solo. In BeyondBots, we classify the motive at the *cluster level* rather than the *post level*, boosting detection robustness.
- **Mukherjee et al. (2013)** (Analysis of Yelp Fake Reviews Filter): Uncovers that paid campaigns maintain zero negativity. In Stage 4, we compute a 'criticism-absence score' (uniform positive sentiment polarity) to isolate commercially bought reviews.
- **Subrahmanian et al. (2016)** (The DARPA Twitter Bot Challenge): Demonstrates that integrating lexical, temporal, and network structures outperforms single-feature models. BeyondBots follows this by combining temporal burst Z-scores, near-duplicate text similarity, and network properties in Stage 3 and Stage 4.
- **Jha et al. (2022)** (Analyzing Code-Mixed Fandom Language): Identifies typical linguistics in Tollywood/Bollywood Twitter (star titles like 'Thalapathy', 'Rebel Star'). This provides the vocabulary baseline for our fan markers lexicon in `config.yaml`.

Table 1. Literature Survey & Methodological Mapping

Paper / Reference	Key Methodology	Core Finding	Direct Mapping to BeyondBots Pipeline
Ratnakar & Kumar (2020)	Vocabulary Analysis	PR spam uses generic superlatives	Stage 4: Stylistic marketing vocabulary score
Ferrara et al. (2016)	CIB Social Bot Theory	Coordinated accounts are human-run	Stage 3: Group near-duplicate text clustering
Benevenuto et al. (2010)	Network Analysis	Spammers show dense interaction topology	Stage 3: Interaction network mappings
Khanuja et al. (2021)	MuRIL Transformers	Pretrained representations for Hinglish/Tamil	Stage 2: Code-mixed dense representations
Campello et al. (2013)	HDBSCAN Clustering	Handles noise outlier classification	Stage 3: Outlier separation from CIB groups
Traag et al. (2019)	Leiden Community	Extracts well-connected subgraphs	Stage 3: Mapping fan-club reply networks
Lundberg & Lee (2017)	SHAP explainers	Local surrogates reveal classification weights	Stage 7: Interactive SHAP importance charts
Pennebaker et al. (2015)	Stylometric Engineering	Pronouns/titles reveal coordination	Stage 4: Star-honorific and pronoun ratios
Blei et al. (2003)	LDA Topic Model	Unsupervised thematic mapping	Stage 5: Cluster keyword/topic summaries
Mukherjee et al. (2013)	Sentiment Modeling	Paid campaigns show zero negativity	Stage 4: Criticism absence score (sentiment)
Jha et al. (2022)	Fandom Lexicons	Star honorifics reveal coordinated fan bases	Stage 4: Dynamic fan marker extraction

2. Methodology & System Architecture

BeyondBots implements an end-to-end 8-stage text mining architecture. The system processes raw review streams and categorizes them into Organic reviews, Coordinated Fan campaigns, or Paid Commercial PR. Below we detail the operational architecture of each stage:

Stage 0: Ingestion & Collection

The pipeline ingests raw social media reviews from files (.csv, .pdf, .docx, .txt, .doc) or retrieves them from the X API v2. If ingestion maps a multi-film dataset, it filters data dynamically by the movie's title. If 0 records match, the collector triggers a simulated fallback data generator to prevent pipeline stoppage.

Stage 1: Preprocessing & Cleaning

Performs text cleanup, removing URLs, usernames, special emojis, and brackets while preserving code-mixed characters. It generates tokenized and lemmatized textual representations using regular expressions for code-mixed terms.

Stage 2: Representation Vectorization

Constructs dual text representation vectors: (1) Sparse TF-IDF matrices to capture interpretable n-gram features (Stage 4), and (2) Dense contextual semantic embeddings using the google/muril-base-cased transformer (Stage 3).

Stage 3: Coordinated Clustering (HDBSCAN + Z-Score)

Finds coordinated groups using semantic duplicates. Dense vectors are clustered using HDBSCAN to group posts with high cosine similarity. It computes a temporal burst score (Z-score) on posting times to flag time-coordinated bursts, and constructs account networks to isolate tight-knit reply communities.

Stage 6: Sentiment Profiling

Executes sentiment analysis to calculate polarity scores. In mock mode, a heuristic model evaluates polarity; in full mode, a RoBERTa model handles multilingual sentiment. These feeds into the criticism-absence feature in Stage 4.

Stage 5: Topic Modeling

Computes cluster-level summaries. For each HDBSCAN cluster, a centroid-based topic extraction technique computes tf-idf keywords to identify the campaign's talking points.

Stage 4: Feature Engineering

Translates raw data into four normalized linguistic metrics at the cluster level: **(a) Marketing Copy Score:** generic superlatives and promo keywords; **(b) Fan Identity Score:** star honorific titles and fandom tags; **(c) Detail Specificity Score:** mentions of specific scenes, plot details, or actors; **(d) Criticism Absence:** uniformity of positive sentiment polarity.

Stage 7: Classification & SHAP Surrogate Explainer

A zero-shot prompt with few-shot examples feeds the cluster metrics, keyword topics, and text samples into Claude-3-5-Sonnet to classify the cluster's motive (organic, fan-coordinated, or paid-commercial). A Logistic Regression surrogate model is fit on the features against LLM predictions to calculate SHAP feature weights for UI explainability.

Stage 8: Case Study Validation

Flagged clusters are automatically cross-checked against documented fan-wars and paid-review scandals stored in config.yaml to verify the classifier's precision against real-world media events.

3. Technology Stack & Frameworks

The project is developed using a professional, modern Python data science stack. The primary packages and frameworks used include:

- **Python 3.11:** The core programming language chosen for strong library support in NLP and ML.
- **Streamlit (v1.61.1):** Used to construct the web application dashboard with responsive controls and page steppers.
- **Plotly (v6.9.0):** Provides high-fidelity, interactive, dark-themed visualizations (donut charts, timelines, PCA projections, and SHAP bar charts).
- **PyTorch & Transformers (Hugging Face):** Used for running dense text representations with MuRIL (Multilingual Representations for Indian Languages) and sentiment analysis.
- **Scikit-Learn:** Handles sparse TF-IDF vectorization, Logistic Regression surrogate models, and PCA dimensionality reduction.
- **Pandas & Numpy:** Forms the foundation for data wrangling, mathematical manipulations, and data frame parsing.
- **SHAP (SHapley Additive exPlanations):** Provides game-theoretic model explanations for surrogate classifier validation.